

Motivation

Transimpedance amplifiers when coupled with a photodetector are a form of interconnect between photonic and electronic world - they offer a way to change signal domain from optical to electrical.

The poster presents the design procedure for a variable-gain transimpedance stage, using a Python script for automatic selection of the compensating capacitor for a given set of circuit parameters.

Laser Absorption Spectroscopy

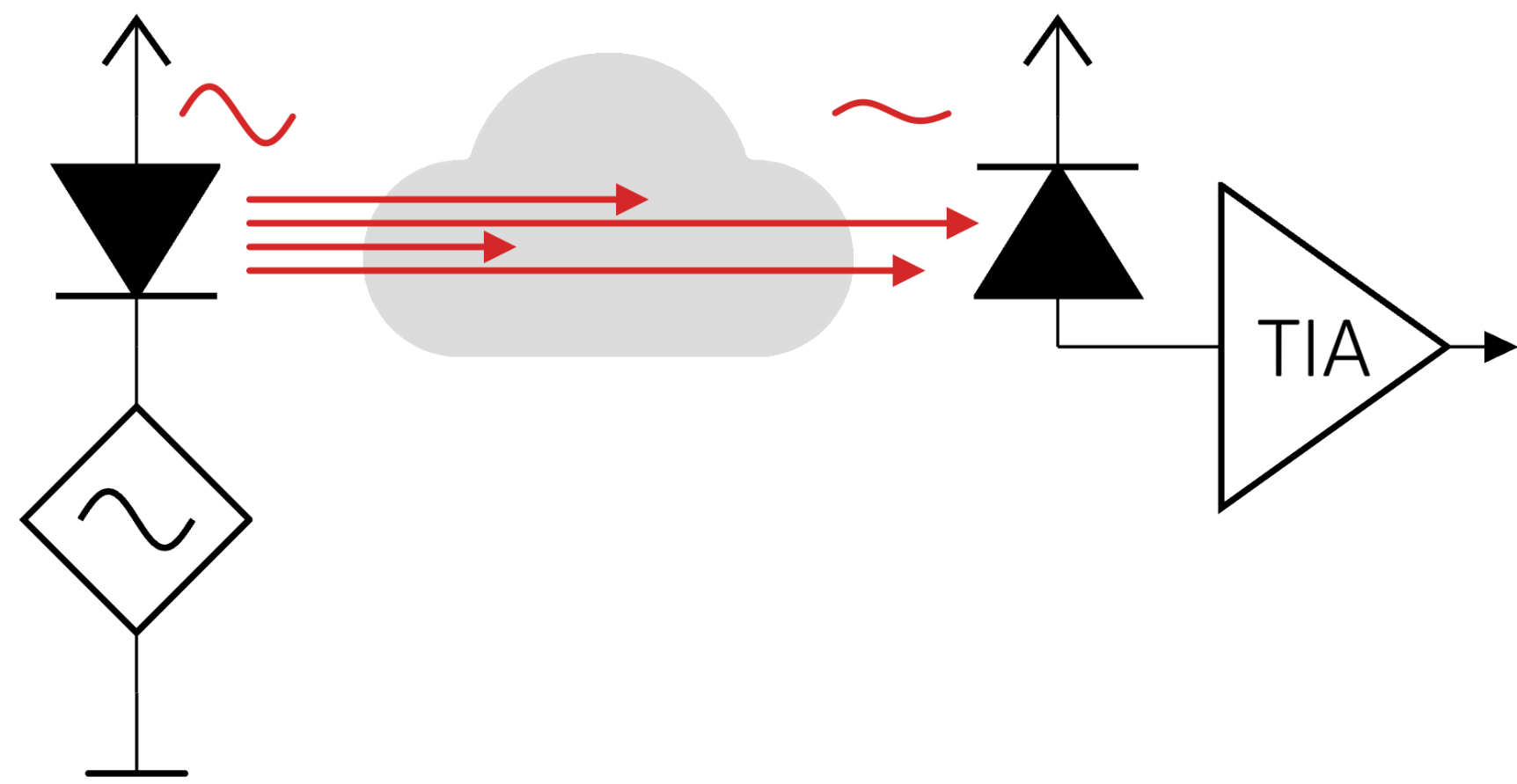


Figure 1: Typical laser absorption spectroscopy system

Laser absorption spectroscopy (LAS) is a family of techniques for gas analysis and sensing. Typical LAS system consists of laser and its modulator, optical path and photodiode coupled with readout electronics [1].

Transimpedance amplifier topology

Designing and optimizing an analog amplifier gets very complicated, even for circuits with only a few components. That is why some designers turn to automatic design algorithms [2] to shorten the design time and maximize circuit's performance.

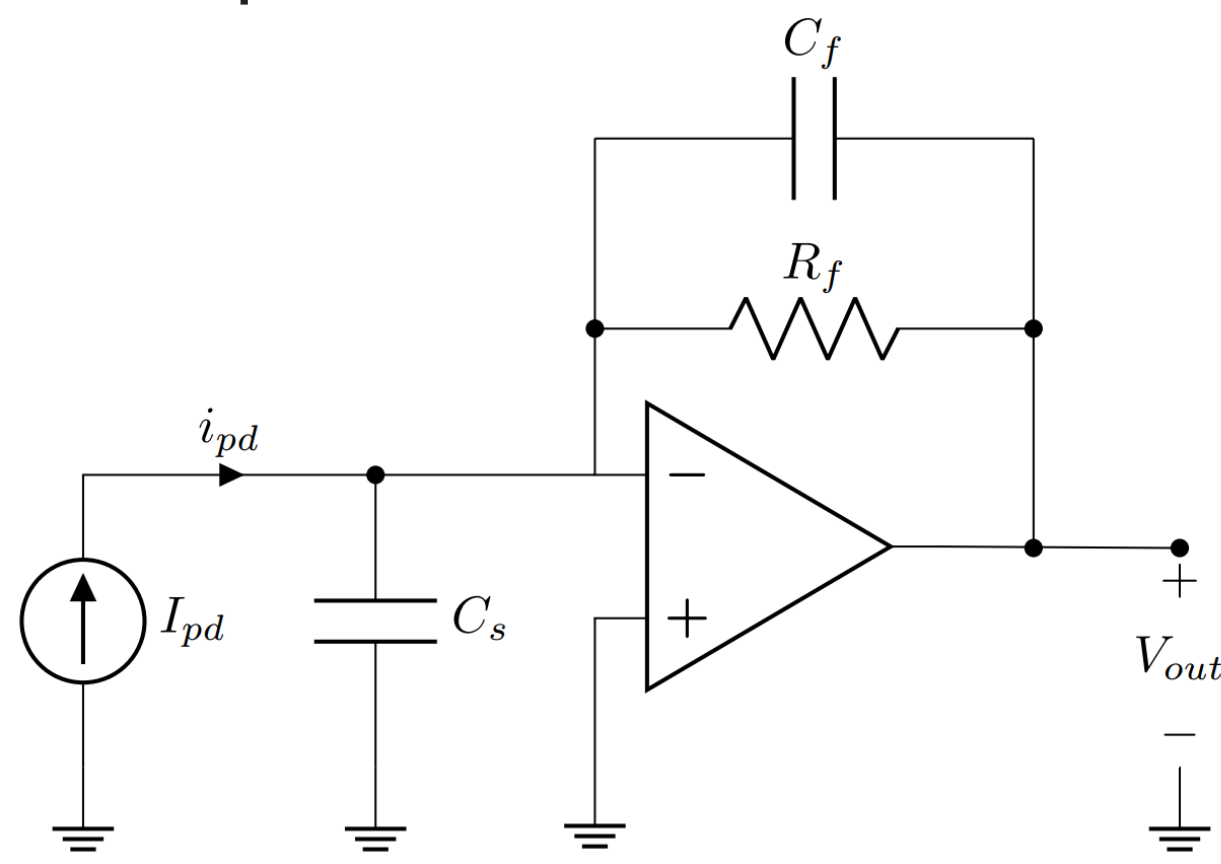


Figure 2: Compensated transimpedance amplifier topology schematic

Compensated transimpedance amplifier system can be simplified to a second degree system with two poles and one zero (Eq. 1).

By modifying C_f we can change the location of the system's zero and alter the phase behaviour. This unfortunately also reduces bandwidth of our amplifier.

$$\beta(s) \cdot A_{OL}(s) = \frac{1 + sC_f R_f}{1 + s(C_s + C_f)R_f} \cdot \frac{A_0}{1 + s \frac{A_0}{2\pi GBW}} \quad (1)$$

Amplifiers usually work in a closed loop system. Its transmittance is given by this formula:

$$H(s) = \frac{A(s)}{1 + A(s) \cdot \beta(s)}$$

Barkhausen stability criterion can be applied to check if a system exhibits oscillatory behaviour:

$$|A\beta| = 1$$

$$\angle A\beta = -180^\circ$$

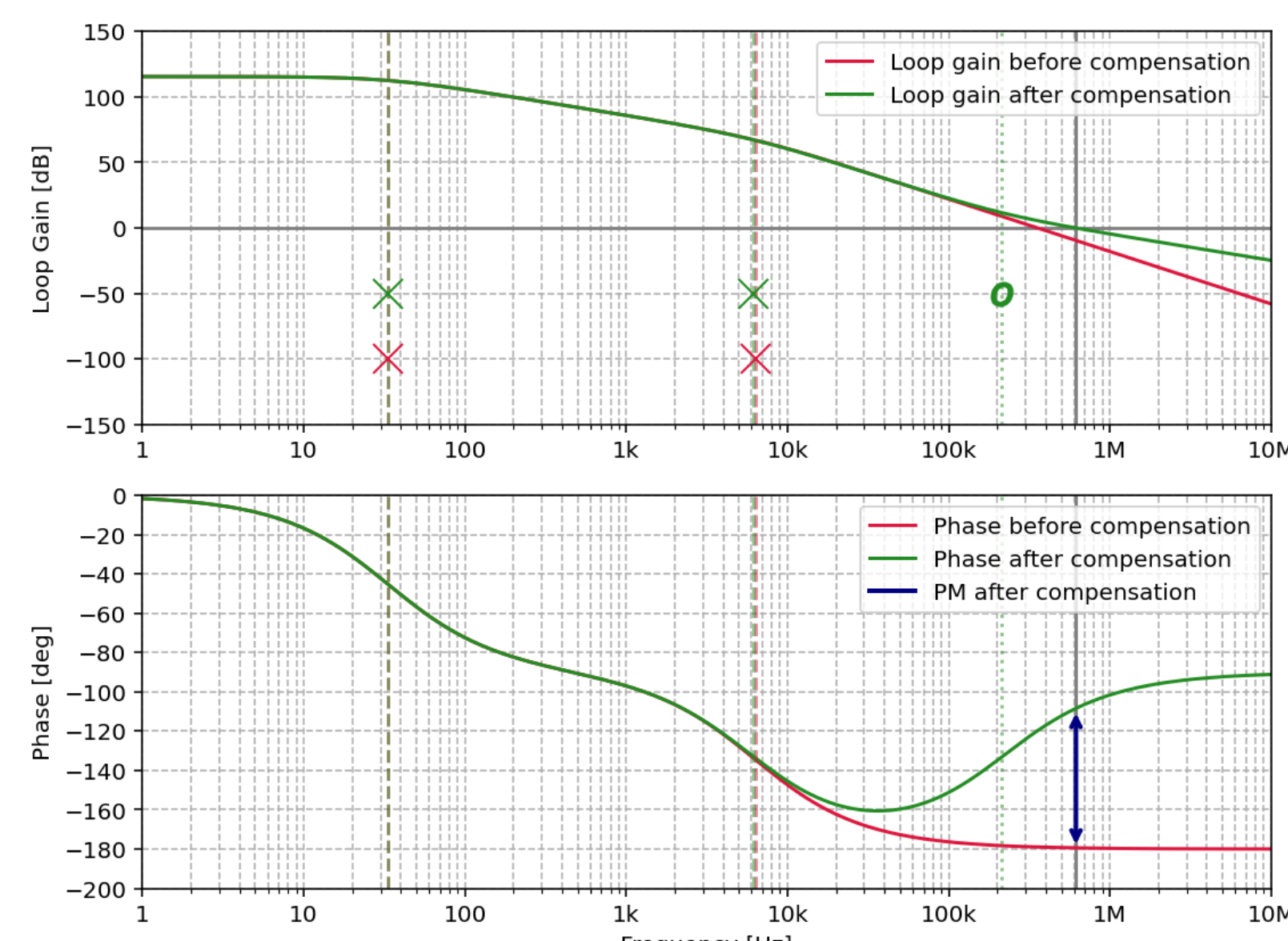


Figure 3: Loop-gain Bode plots of an unstable (red) and stable (green) systems.

Design Procedure

```
import pyTIAModule as tia

# OA parameters
Ao_db = 115
GBW = 20e6

# photodiode parameters
Cs = 35e-9

# feedback parameters
Rf_arr = [20_000] # ohm
Cf_min = 470e-15 # 470fF
Cf_max = 10e-6 # 10uF

# change dB to lin gain
Ao_lin = tia.Ao_db_to_lin(Ao_db)

# generate capacitors based on the available series
# E3(20\%) E6(20\%) E12(10\%) E24(5\%) E48(2\%) E96(1\%) E192(0.5\%, 0.25\%, 0.1\%)
e96_base = tia.generate_e_series_bases(96)
Cf_list = tia.generate_capacitors(e96_base, Cf_min, Cf_max)

f, s = tia.generate_f_vector(fmin=1, fmax=1e9, decpoints=20) # 1Hz to 1GHz
tia.find_best_Cf(f, s, Ao_lin, GBW, Cs, Rf_arr, Cf_list, target_PM=60)
```

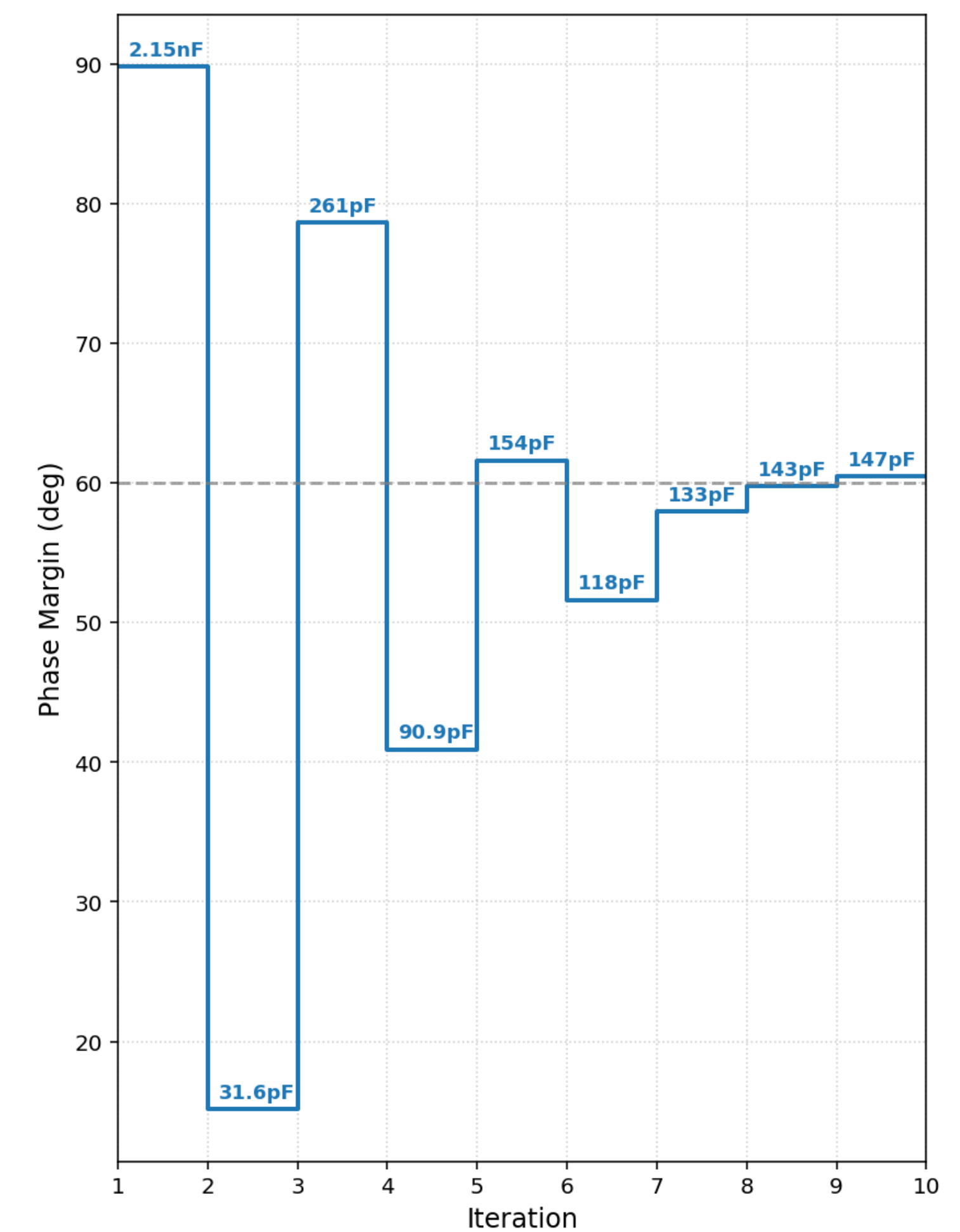


Figure 4: Resulting binary search

SPICE validation

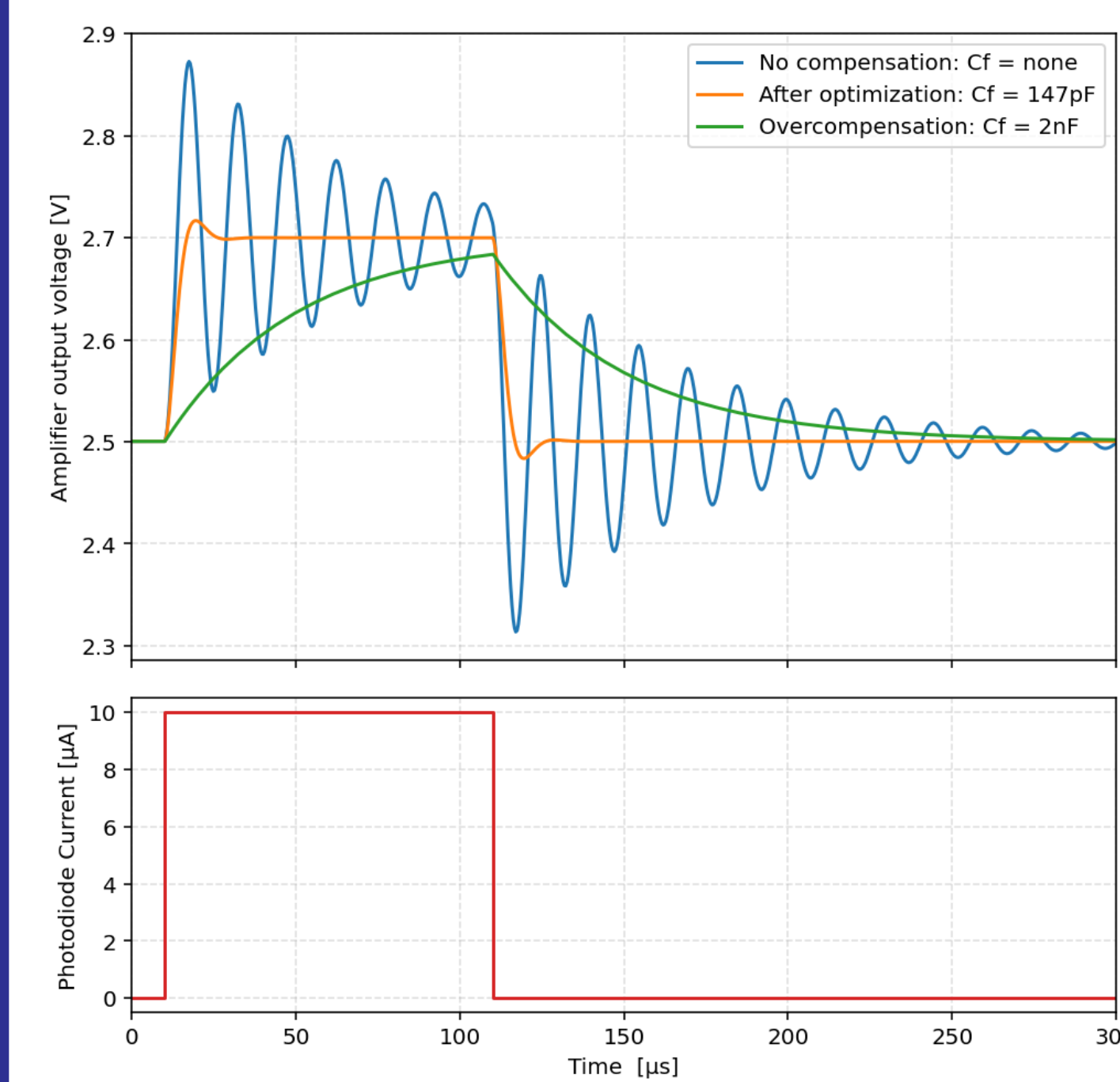


Figure 5: SPICE simulations results for different compensation capacitor values

The circuit presented in the **Transimpedance Amplifier Topology** section was simulated in SPICE environment for three capacitor values: 1 fF (emulating no capacitance), 147 pF (cap given by the optimization algorithm) and 2 nF.

Significant ringing is visible for the simulation step with no compensation capacitor, while the 2 nF capacitor "slows" the circuit too much.

Circuit optimized for $PM=60^\circ$ provides us with the fastest possible settling time with minimal overshoot.

Summary

The presented methodology was used to design a variable-gain TIA (Fig. 6) with six gain settings for methane LAS readout electronics. The proposed automatic compensation method greatly reduced design time by optimizing the circuit for maximum performance for each setting.

Furthermore, this methodology can be integrated with other lookup-table-based approaches [3] (such as the g_m/I_D technique) and Python-compatible tools (pygmid) to further automate the design process.

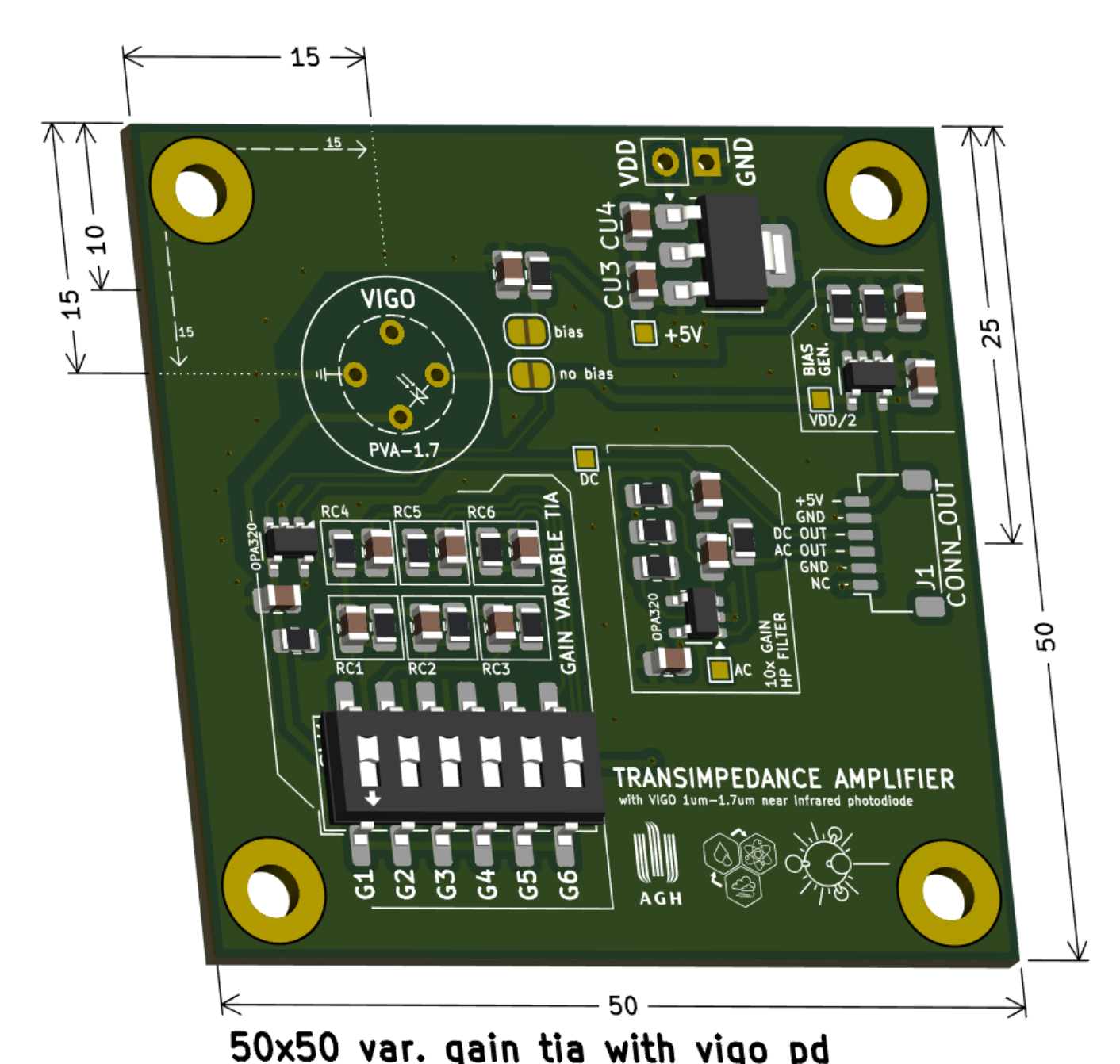
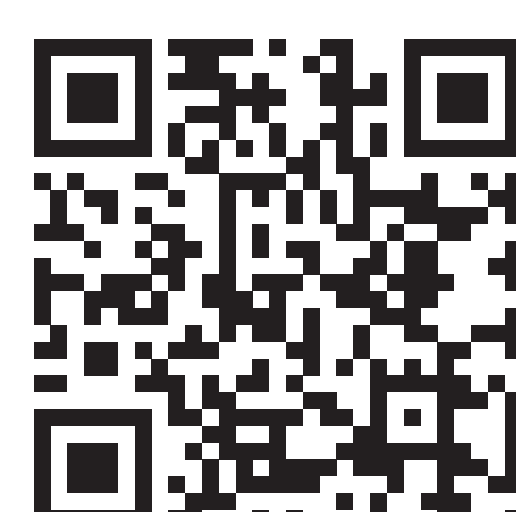


Figure 6: Designed electronics for laser gas spectroscopy application

REFERENCES

- [1] VIGO Photonics S.A., "Infrared detectors for spectroscopy." Applications, 2022. Accessed: 2026-05-18.
- [2] P. Kaczmarczyk, "Automatic design of a cascoded-inverter-based charge-sensitive amplifier using gm/id technique and particle swarm optimization in 28 nm cmos," 2023.
- [3] P. G. A. Jespers and B. Murmann, *Systematic Design of Analog CMOS Circuits: Using Pre-Computed Lookup Tables*. 2017.

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pyTIA
tool for automatic selection
of compensating capacitor
in transimpedance amplifier
circuits

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